Piecewise Linear Regression

Optimization problem:

$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

with data pairs $\{(a_i, b_i)\}_{i=1}^{n}$.

Equivalent linear programming:

$$\min_{x, t_1, ..., t_n} \quad \sum_{i=1}^{n} t_i$$

$$\text{s.t.} \quad -t_i \le a_i x - b_i \le t_i, \quad \forall i$$

Piecewise Linear Regression

How to minimize $\sum_{i=1}^{n} |a_i x - b_i|$ without using linear programming?

Piecewise Linear Regression

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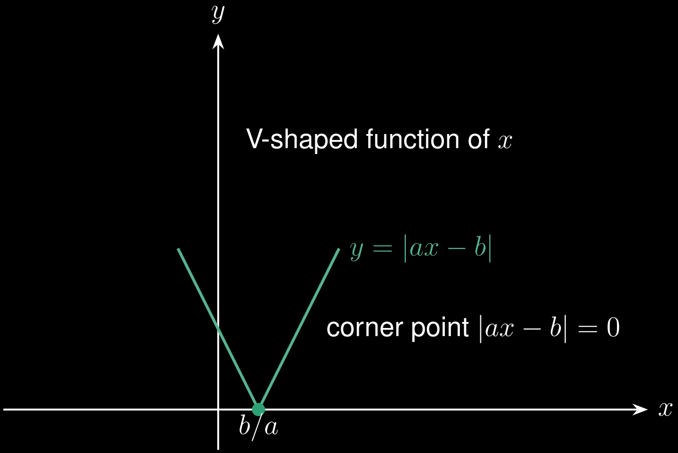

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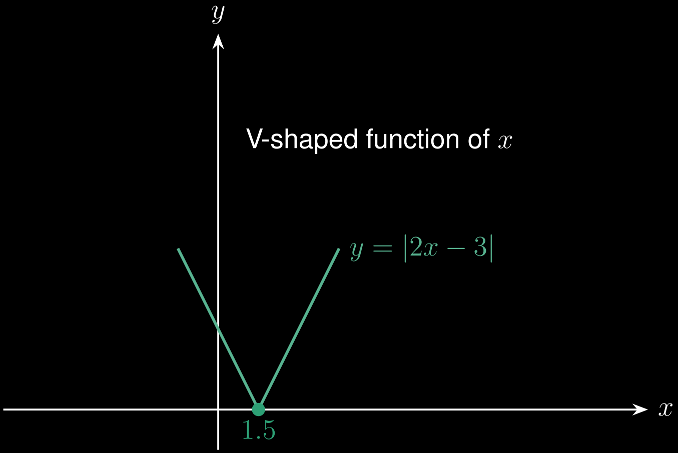

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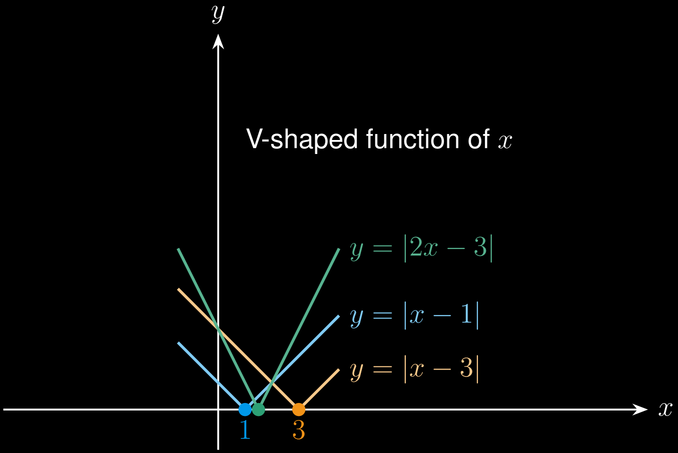

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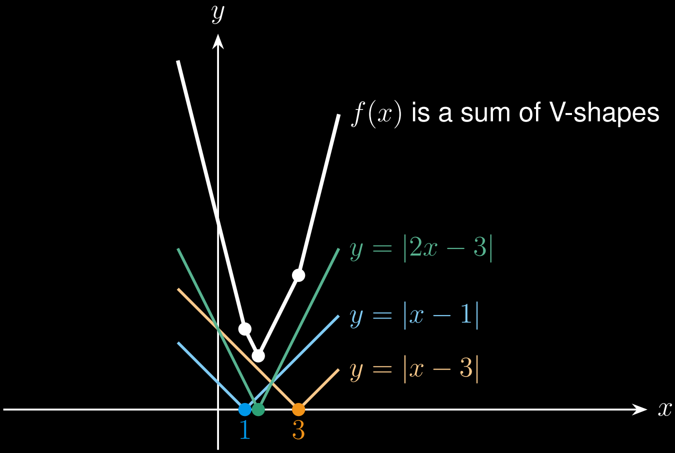

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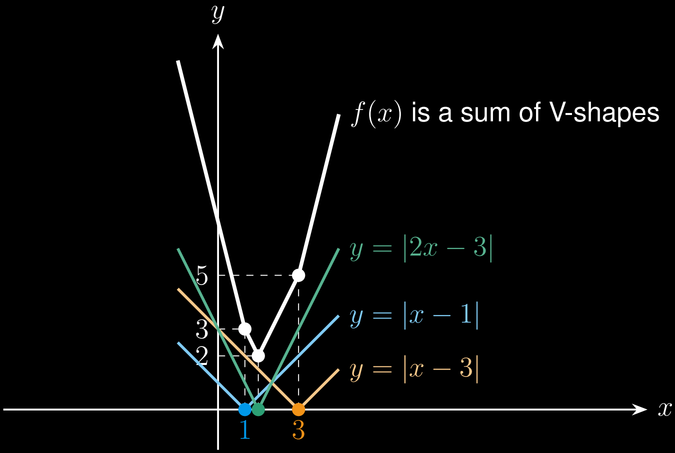

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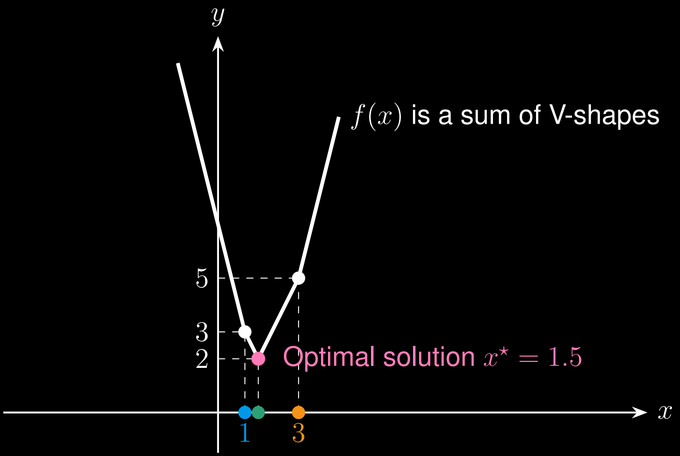

Thanks for your attention!

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